

Total No. of Questions : 8]

SEAT No. :

PC2809

[Total No. of Pages : 3

[6352]-33

S.E. (Electronics and Computer Engineering)

DATA STRUCTURES AND ALGORITHMS

(2019 Pattern) (Semester - III) (204184)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.
- 4) Use of non-programmable scientific calculator is allowed.

- Q1)** a) What is stack data structure? Write a C structure for a stack of integer implemented using array. Explain stack operations. [6]
b) Write an algorithm for evaluating a postfix expression using stack. [6]
c) Evaluate following postfix expression with the help of stack. [5]
i) $623 + - 382 / + * 2 \$ 3 +$
ii) $231 * + 9 -$

OR

- Q2)** a) Write the following C functions when the stack is implemented using array of integers. [6]
i) is empty
ii) is full
iii) Push
iv) Pop
b) Compare stacks and queues. [6]
c) What is queue? Explain basic operations on queue. State queue empty and full conditions when queue is implemented using array. [5]
- Q3)** a) Compare SLL with DLL? [6]
b) Write a C declaration for a structure of - [6]
i) Singly Linked List
ii) Doubly Linked List
c) Differentiate between static and dynamic memory allocation. Name and explain in brief dynamic memory allocation functions. [6]

OR

P.T.O.

Q4) a) Compare array and linked list. Declare in 'C' a node structure for doubly linked list. [6]

b) Write a function in C to delete a node in SLL. [6]

c) Represent the following polynomials using linked list. [6]

i) $25x^5 + 15x^4 + 5x^3 + 2x^2 + 100$

ii) $16x^8 + 9x^4 + 5x^2 + 2x$

Q5) a) Define the following terms with respect to tree data structure with suitable example. [6]

i) Degree of a node

ii) Height of a node

iii) Height of tree

b) Construct a binary search tree for the following set of elements. [6]

50, 25, 85, 45, 15, 95, 75

Traverse the resulting binary search tree in post-order manner.

c) What is threaded binary search tree. Explain with suitable example. [5]

OR

Q6) a) From the given traversal construct the binary tree. [6]

Inorder : D B F E A G C L J H K

Postorder : D F E B G L J K H C A

b) Explain with suitable example how binary tree can be represented using [6]

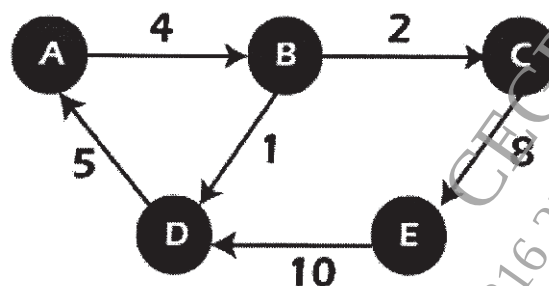
i) Array

ii) Linked List

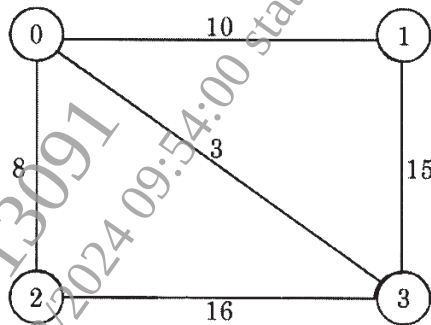
c) Construct a threaded binary search tree for the following data. [5]

100, 50, 200, 300, 20, 150, 70, 180, 120, 30

Q7) a) What do you mean by adjacency matrix and adjacency list. Give the adjacency matrix and adjacency list of the following graph. [6]

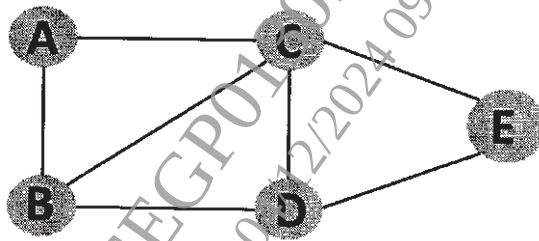


- b) Construct a minimum spanning tree for the following graph using Kruskal's Algorithm. [6]



- c) Explain Dijkstra's shortest path algorithm with suitable example. [6]
OR

- Q8)** a) Write an algorithm for depth first search technique. [6]
b) Perform depth first search algorithm on the following graph. [6]



- c) Define graph data structure. Explain following graph terms with illustrations/diagrams. [6]
i) Complete graph
ii) Connected graph
iii) Subgraph

